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PUSL3190 Computing Project

Abstract

AI Sign Language Speech Translator

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Abstract

For years, good communication between the deaf and hard-of-hearing and the broader hearing community has been hampered by the absence of readily available technology for translating sign language in real time. Sign language is the preferred mode of linguistic expression for around 70 million deaf people worldwide, but has remained largely unknown to the hearing public, thus perpetuating a communication chasm that has serious social implications. Previous literature has proposed automated sign language recognition systems that have been limited significantly by the need for specialized depth sensing hardware, instrumented sensor gloves, or cloud-based inference pipelines that are vulnerable to network latency and connectivity failure. Such systems are not practical for deployment in everyday assistive contexts. To address these limitations, we propose SignaSpeech, a mobile-based, real-time sign language recognition and speech conversion system that can perform fully on-device deep learning inference on commodity Android smartphones without specialized peripheral equipment. The core recognition model used was the YOLOv8 object detection architecture in the PyTorch framework through the Ultralytics library. We trained on a custom annotated dataset of approximately 7,000 images, annotated using the Roboflow annotation platform and exported in YOLO format, representing 8 gesture categories that are commonly used signs. One basic design principal of the proposed system was extensibility and support for the entire American Sign Language (ASL) alphabet is planned for future development phases. The trained model was converted to TensorFlow Lite format and embedded into a Flutter-based Android application that takes live YUV camera frames, performs real-time color space conversion, resizes, and normalizes each frame into a 224×224-pixel input tensor, performs on-device inference, and outputs audible speech via a Text-to-Speech engine, all without network connectivity. Evaluation on a held-out test partition resulted in an overall gesture classification accuracy of 91.3% and a mean end-to-end inference latency of 41.4 milliseconds on mid-range Android hardware, confirming the viability of SignaSpeech as a scalable and accessible solution for bridging the communication divide between deaf and hearing communities.

Keywords — Real-time detection, YOLOv8, mobile application, Flutter, deep learning, computer vision, sign language recognition, TensorFlow Lite